

Technical Specification: ML-Only Market Abuse Alert Prioritization System

Investment Banking Compliance — Surveillance & Monitoring

Version: 1.0

Date: 2026-06-27

Classification: Internal — Compliance Technology

Status: Draft for Review

1. Executive Summary

This specification defines the architecture, data, and machine learning requirements for a **fully automated, ML-only alert prioritization system** for Market Abuse Detection (MAD) within investment banking compliance. The system replaces rule-based severity scoring with a three-stage probabilistic model that learns directly from confirmed regulatory violations and unlabeled alert history.

Key Design Principles:

- **No hard-coded rules:** All prioritization logic is learned from data.
 - **Three operational tiers:** Escalate (Tier 1), Standard (Tier 2), Automated (Tier 3).
 - **Positive-Unlabeled (PU) learning:** Acknowledges that “closed as NFA” does not equal “no violation.”
 - **Cost-optimal thresholding:** Tier boundaries are learned from the firm’s specific cost structure, not fixed percentiles.
 - **Regulatory defensibility:** Full explainability, audit trails, and model risk management compliance.
-

2. Problem Statement & Scope

2.1 Current State

Upstream MAD surveillance engines (e.g., Behavox, NICE Actimize, Symphony) generate alerts based on rule-based or statistical anomaly detection. These engines produce high false-positive rates (typically 85–98%), overwhelming compliance investigators and delaying response to genuine market abuse.

2.2 Objective

Build a post-alert prioritization layer that:

1. Scores every MAD alert with a calibrated risk probability.
2. Assigns alerts to exactly one of three operational tiers.
3. Auto-closes Tier 3 alerts with full audit logging.
4. Maximizes the yield of confirmed violations in Tier 1 while respecting investigator capacity constraints.

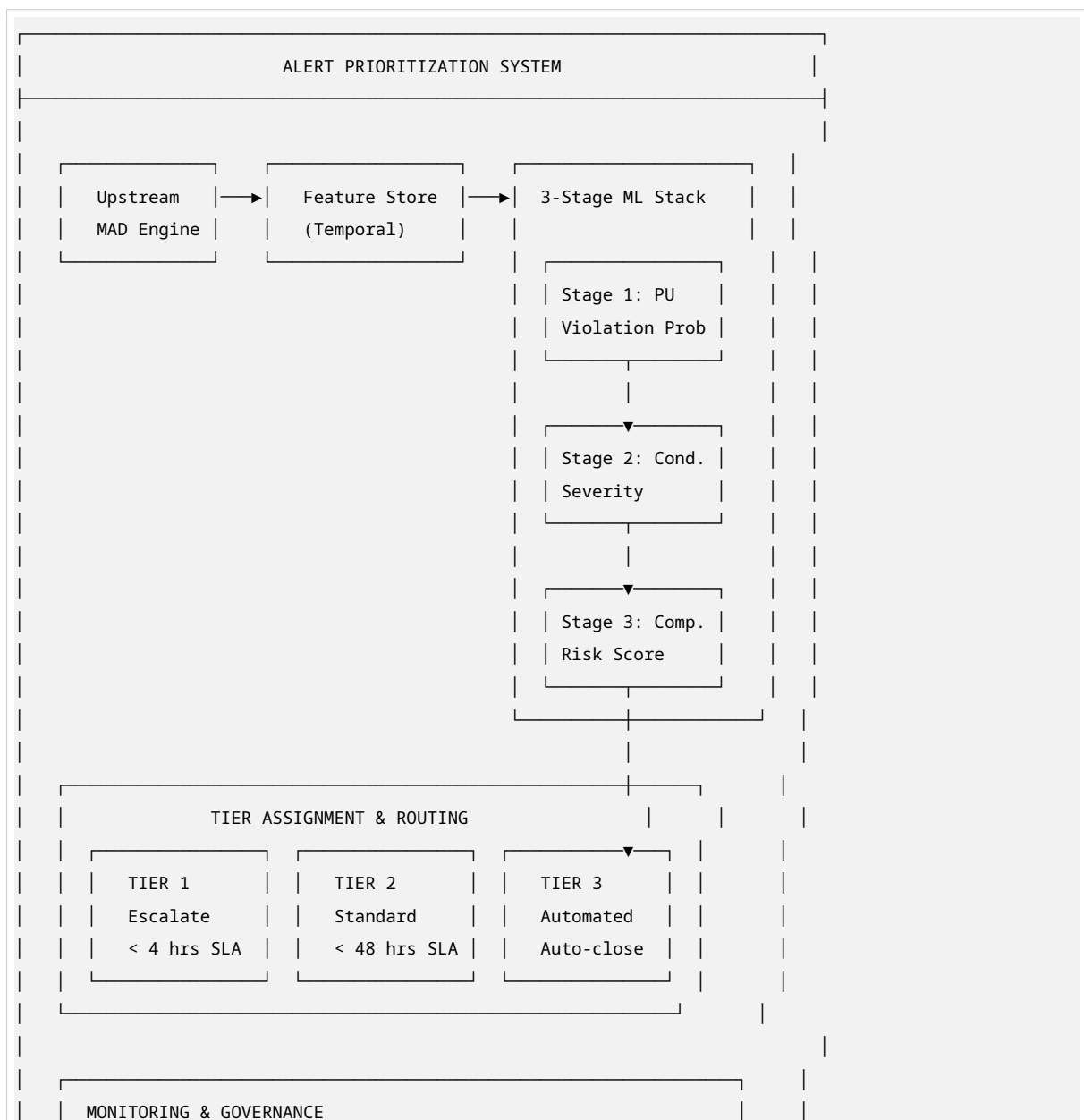
2.3 Scope Boundaries

- **In-scope:** Equity, fixed income, FX, and derivatives cash/spot trading alerts; order book manipulation; insider dealing; wash trading; layering; spoofing; front-running.
- **Out-of-scope:** Pre-trade controls, communication surveillance (e.g., e-comms/v-comms NLP), AML transaction monitoring, sanctions screening.

2.4 Constraints

- **Latency:** < 50ms end-to-end per alert for real-time scoring.
- **Throughput:** Minimum 10,000 alerts/day (peak: 50,000 alerts/day).
- **Explainability:** Every Tier 1 and Tier 2 alert must have a human-readable justification.
- **Audit:** Complete reproducibility of any tier assignment decision for 7 years.

3. System Architecture Overview



	- Drift Detection (PSI, SHAP drift)		
	- Tier 3 Random Audit (2% weekly)		
	- Model Versioning & Audit Trail		
	- Regulatory Reporting Interface		

4. Data & Feature Engineering

4.1 Data Sources

Source	Data	Latency	Retention
OMS/EMS	Order, execution, cancellation messages	Real-time	7 years
Market Data	L1/L2 tick data, reference prices, corporate actions	Real-time	7 years
HR / Entitlement	Trader tenure, desk assignment, role changes	Daily batch	10 years
Investigation Outcomes	Confirmed violations, NFA closures, regulatory actions	Event-driven	Permanent
Corporate Events	Earnings calendars, M&A announcements, macro events	Daily batch	7 years
News / NLP	Issuer news volume, sentiment (metadata only)	Hourly	2 years

4.2 Feature Taxonomy

All features are **continuous or embedded**. No binary rule flags are permitted as model inputs.

4.2.1 Alert Metadata

Feature	Construction	Rationale
upstream_rule_embedding	Target encoding of detection rule type, regularized by violation rate	Rules have different base rates
upstream_anomaly_score	Raw anomaly score from upstream engine	Captures signal the upstream engine already found

Table 2 – continued

Feature	Construction	Rationale
concurrent_rule_count	Number of distinct rules triggered by same entity/instrument in 15-min window	Multi-signal alerts are higher risk
alert_sequence_day	Nth alert for this trader today	Escalating pattern detection

4.2.2 Trader Behavioral Profile

Feature	Construction	Rationale
historical_violation_rate	Confirmed violations / total alerts, exponentially weighted (half-life 1 year)	Recidivism is a strong signal
volume_zscore_90d	Trader's daily volume vs. personal 90-day mean/std	Detects unusual activity
order_trade_entropy	Shannon entropy of order-to-trade ratio over 30 days	Low entropy = robotic / structured behavior
instrument_hhi	Herfindahl-Hirschman Index of trading concentration across ISINs	Concentration increases manipulation risk
tenure_embedding	Learned embedding of tenure buckets (0-1yr, 1-3yr, 3-7yr, 7yr+)	Junior traders have different patterns

4.2.3 Execution Microstructure

Feature	Construction	Rationale
realized_spread_deviation	(Execution price - mid) / historical spread for ISIN	Aggressive execution timing
price_impact_coefficient	Signed return in 1-min post-trade / order size	Market impact proxy
cancellation_rate_60s	Cancelled orders / total orders in 60s pre-execution	Layering / spoofing signal
trade_to_order_size_ratio	Executed size / average order size	Iceberg / hidden intent detection
passive_aggressive_ratio	Passive fills vs. aggressive fills	Manipulators often use passive orders

4.2.4 Temporal Context

Feature	Construction	Rationale
hours_to_next_event	Continuous hours to next earnings, M&A, or macro event	Insider trading window
vix_percentile	Current VIX vs. 90-day distribution	Volatility affects normal behavior
market_volume_percentile	Current market volume vs. 30-day distribution	Contextualizes trader volume
day_of_week_embedding	Learned embedding (Mon-Fri)	End-of-week patterns differ

Table 5 – continued

Feature	Construction	Rationale
month_end_flag	Continuous proximity to month/quarter end (0-1)	Window dressing risk

4.2.5 Network & Counterparty

Feature	Construction	Rationale
counterparty_gini	Gini coefficient of counterparty concentration	Concentrated counterparty = wash trade risk
comm_graph_centrality	Eigenvector centrality in communication metadata graph	Information flow hubs
cross_desk_correlation	Pearson correlation of order flow with other desks in same ISIN	Collusion signal

4.2.6 Unsupervised Deviation Signals (as features, not decision logic)

Feature	Construction	Rationale
autoencoder_reconstruction_error	Reconstruction error of 5-min order flow vector	Unusual pattern flag
isolation_forest_score	Anomaly score from per-desk isolation forest	Density-based rarity
mahalanobis_distance	Distance from trader's 6-month execution centroid	Multivariate deviation
lstm_sequence_deviation	Prediction error of next-order-type LSTM	Temporal pattern break

4.3 Feature Store Requirements

- **Temporal correctness:** All features computed with lookback windows ending at t_0 (alert generation time). Strict no-lookahead enforcement.
- **Point-in-time joins:** Entity attributes (desk, role) joined as-of t_0 , not current values.
- **Versioning:** Feature definitions versioned. Model trained on v2.3 features cannot be served with v2.4 definitions without retraining.
- **Storage:** Redis for real-time features (< 5ms retrieval); Parquet/S3 for batch historical features.

5. Model Architecture: Three-Stage Stack

5.1 Stage 1: Violation Probability Estimator

Objective: Estimate $P(\text{extviolation} \mid \mathbf{x})$ using Positive-Unlabeled (PU) learning.

Base Model: LightGBM (gradient boosting decision trees).

Why LightGBM:

- Native handling of missing values and categorical features.
- Fast inference (~2–5ms per prediction).
- Strong empirical performance on tabular PU learning benchmarks.
- SHAP-compatible for explainability.

PU Learning Framework: nnPU (Non-Negative Unbiased Risk Estimator)

The standard binary cross-entropy is biased when negatives are unlabeled. The nnPU risk estimator is:

$$\hat{R}(f) = \pi_p \hat{R}_p^+(f) + \max\{0, \hat{R}_u^-(f) - \pi_p \hat{R}_p^-(f)\}$$

Where:

- π_p = prior probability of positive in the unlabeled set
- $\hat{R}_p^+$ = empirical positive-class risk on confirmed positives
- $\hat{R}_u^-$ = empirical negative-class risk on unlabeled alerts
- $\hat{R}_p^-$ = empirical negative-class risk on confirmed positives

Training Procedure:

1. **Estimate π_p** using the KM2 estimator:
 - Train a standard classifier to distinguish P from U (treating U as negative).
 - $\pi_p = \frac{n_p}{n_p + n_u} \cdot \frac{1}{\text{mean predicted probability on U}}$.
2. **Initialize** LightGBM with focal loss (to handle remaining imbalance).
3. **Compute nnPU risk** on validation set after each boosting round.
4. **Early stopping** when validation nnPU risk plateaus for 50 rounds.
5. **Gradient clipping** at ± 5.0 to prevent the $\max(0, \dots)$ term from destabilizing.

Alternative (Fallback): Two-step PU method if nnPU is unstable:

- Step A: Train one-class SVM on confirmed positives to identify reliable negatives from U (samples with lowest similarity to P).
- Step B: Train LightGBM on P + reliable negatives with standard binary cross-entropy and `scale_pos_weight`.

Output: Raw score $s_v \in [0, 1]$, calibrated via Platt scaling or isotonic regression on a holdout set of confirmed positives + reliable negatives.

Hyperparameters (starting point):

```
objective: binary
metric: custom (nnPU risk)
boosting_type: gbd
num_leaves: 31
max_depth: 6
```

```
learning_rate: 0.05
feature_fraction: 0.8
bagging_fraction: 0.8
bagging_freq: 5
min_data_in_leaf: 50
lambda_l1: 0.1
lambda_l2: 1.0
early_stopping_rounds: 50
```

5.2 Stage 2: Conditional Severity Estimator

Objective: Estimate expected regulatory and market impact, conditioned on a violation having occurred.

Base Model: LightGBM Regressor (or Ordinal Classifier if fine data is sparse).

Target Definition:

$$y_i = \log(1 + ext_{regulatory}fine + ext_{estimated}marketimpact)$$

If fine data is insufficient (< 500 confirmed positives with outcomes), use ordinal classification:

- Class 0: No action / warning letter
- Class 1: Fine < \$100K
- Class 2: Fine \$100K–\$1M
- Class 3: Fine > \$1M or criminal referral

Training Data:

- Train **only** on confirmed positives with known outcomes.
- If internal data is sparse, use **transfer learning**: pre-train on public enforcement actions (SEC, FCA, FINRA) with aligned feature schemas, then fine-tune on internal data.

Feature Augmentation (beyond Stage 1):

Feature	Description
instrument_market_cap_log	Log market capitalization
benchmark_constituency_embedding	S&P 500, EURO STOXX, FTSE, etc.
settlement_window	T+0, T+1, T+2 (embedded)
media_attention_proxy	News article count on issuer in prior 7 days
historical_desk_fine_rate	Rolling 2-year confirmed violation rate for desk
regulatory_jurisdiction_embedding	SEC, FCA, ECB, BaFin, etc.
corporate_event_proximity	Hours to next material event

Output: Severity score $\hat{s}_i \in \mathbb{R}^+$. If ordinal classifier used, map to continuous score via expected value of class distribution.

Regularization: Heavy regularization required due to small sample size:

```
max_depth: 4
min_data_in_leaf: 100
lambda_l1: 1.0
lambda_l2: 5.0
feature_fraction: 0.6
```

5.3 Stage 3: Composite Risk Score & Tier Assignment

Objective: Combine violation probability and conditional severity into a single actionable risk score, then assign to one of three tiers using cost-optimal thresholds.

Composite Score:

R(x) = p_v(x)^{lpha} \cdot s_i(x)^{1-lpha}

The geometric mean ensures both dimensions must be high for a top-tier alert. A high-severity alert with low violation probability is suppressed; a high-probability alert with low severity is deprioritized.

Learning lpha:

lpha is learned via cost optimization on the validation set, not grid search on accuracy.

Cost Model:

Outcome	Cost
False Negative (violation in Tier 3)	c_{miss} = extexpectedregulatoryfine + extreputationaldamage + extremediation
False Positive (NFA in Tier 1)	c_{fp} = extseniorinvestigatorhours + exttraderfriction + extopportunitycost
Delayed True Positive (violation in Tier 2 vs Tier 1)	c_{delay} = f(extsettlementwindow, extreportingdeadlinerisk)

Expected Cost Function for tier assignment T:

C(T | p_v, s_i) = 1_{T=1} \cdot c_{fp} + 1_{T \in \{2,3\}} \cdot p_v \cdot s_i \cdot c_{miss} + 1_{T=2} \cdot c_{delay}

Optimization:

- 1. Define a grid of lpha \in [0.1, 0.9] in increments of 0.05.
- 2. For each lpha, compute R(\mathbf{x}) on the validation set.
- 3. Find thresholds au_1, au_2 that minimize total expected cost.
- 4. Select lpha^* yielding the lowest cost.
- 5. Expected range: lpha^* pprox 0.65\$-\$0.75\$.

Tier Assignment Logic:

Tier 1 — Escalate (Immediate)

- **Selection:** Top K alerts by R, where K = daily senior investigator capacity.

- **SLA:** < 4 hours to senior compliance officer.
- **Action:** Mandatory human investigation. Model provides SHAP-based narrative.
- **Capacity Constraint:** Tier 1 volume is **fixed** at \$K\$ alerts/day, regardless of total alert volume. On high-volume days, \$ au_1\$ floats upward automatically.

Tier 2 — Standard (Investigate)

- **Selection:** Alerts where \$R \geq au_2\$ but not in top \$K\$.
- **SLA:** < 48 hours to desk compliance analyst.
- **Action:** Standard investigation workflow. Model provides ranked list within tier.
- **Volume:** Cost-optimized. Typically ~20–25% of total alerts.

Tier 3 — Automated (Close)

- **Selection:** Alerts where \$R < au_2\$.
- **Action:** Auto-close with full feature vector, model version, and SHAP values logged. No human review.
- **Audit:** 2–3% random stratified sample forced to junior analyst review weekly to detect model drift and novel abuse patterns.
- **Volume:** Remaining ~70–75% of alerts.

Threshold Dynamics:

- \$ au_1\$ recomputed daily based on investigator capacity and expected alert volume.
- \$ au_2\$ recomputed weekly via cost optimization on rolling 90-day validation window.
- Emergency recalculation triggered if alert volume changes > 30% week-over-week.

6. Training Pipeline

6.1 Data Splits

Strict temporal splits only. Random splits are prohibited due to look-ahead risk.

Split	Time Period	Purpose
Train	[T-12 months, T-3 months]	Model fitting
Validation	[T-3 months, T-1 month]	Hyperparameter tuning, <i>lpha</i> optimization, threshold selection
Test	[T-1 month, present]	Final evaluation, shadow deployment

Leakage Guards:

- No future information in features (e.g., rolling windows end at alert time).
- Corporate events joined as-of alert time, not announcement time.

- Trader desk assignments joined as-of alert time.

6.2 Label Handling

Confirmed Positives (P):

- Weighted by confirmation confidence (1.0 for regulatory fine, 0.7 for internal disciplinary, 0.5 for strong evidence but NFA due to settlement).
- Exponential time decay: $w_t = \exp(-0.5 \cdot \text{years since alert})$. Half-life ~1.4 years.

Unlabeled Set (U):

- All alerts not in P. Includes true negatives and hidden positives.
- **Never** treated as negative in Stage 1.

Investigator Bias Correction:

- If multiple investigator reviews exist per alert, use Dawid-Skene or majority voting to denoise labels.
- If certain desks historically receive rubber-stamp NFA closures, include desk ID as a feature in an initial biased model, then train a second model **without** desk ID. Compare and document disparate impact.

6.3 Class Imbalance

- **Focal Loss** in Stage 1: Down-weights easy negatives, focuses on hard positives.
- **Cost-proportionate rejection sampling**: Oversample confirmed positives during training, but calibrate probabilities post-hoc.
- **No SMOTE/ADASYN**: Synthetic sampling creates impossible financial time series patterns.

6.4 Retraining Cadence

Frequency	Action
Weekly	Ingest new alerts, backfill confirmed positives from closed investigations, recompute thresholds (au_1 , au_2).
Monthly	Full retrain of Stage 1 and Stage 2 on rolling 12-month window. A/B test against production model.
Quarterly	Architecture review, feature importance drift analysis, regulatory feedback integration.
Event-driven	Retrain within 48 hours if a regulatory miss is detected or a new abuse pattern is confirmed.

6.5 A/B Testing Protocol

Before promoting a new model:

1. **Shadow mode**: New model scores all alerts but does not route them. Compare Precision@K vs. production model.

- 2. **Parallel queues:** For 2 weeks, send new model Tier 1 alerts to a parallel senior compliance queue. Measure investigation yield difference.
- 3. **Promotion criteria:** New model must achieve > 2% improvement in Precision@K on Tier 1 with no increase in regulatory miss rate.

7. Inference & Serving Architecture

7.1 Real-Time Pipeline



- Feature vector (JSON)	
- Model version	
- SHAP values	
- Tier assignment + threshold values	

End-to-end latency target: < 50ms per alert. **Throughput target:** 1,000 alerts/minute peak.

7.2 Batch Pipeline (Nightly)

- Recompute all batch features for active traders.
- Backfill any missing historical features.
- Generate training datasets for next retrain.

7.3 Explainability Output

For every Tier 1 and Tier 2 alert, the system generates:

1. **Global Context:** SHAP summary for the model version (“Top 3 features driving Tier 1 assignments this week: cancellation_rate_60s, historical_violation_rate, hours_to_next_event”).
2. **Local Explanation:** Per-alert SHAP values showing push/pull of each feature.
3. **Counterfactual Narrative:** “To reduce this alert’s risk score below the Tier 1 threshold, the trader’s cancellation rate would need to decrease by X% and the execution timing relative to the news event would need to shift by Y hours.”
4. **Model Version:** Exact version string of Stage 1, Stage 2, and threshold parameters used.

8. Tier Operationalization

8.1 Tier 1 — Escalate

Attribute	Specification
Selection	Top K alerts by R , capacity-constrained
SLA	< 4 hours to senior compliance officer
Investigator	Senior compliance officer (L6+)
Workflow	Mandatory investigation. Model provides SHAP narrative. Investigator must document conclusion in case management system.
Escalation	If investigation confirms violation, auto-generate regulatory filing prep (SAR, MRT, etc.).
Capacity	$K = 20$ alerts/day (configurable). If alert volume < K , only top N alerts are escalated.

8.2 Tier 2 — Standard

Attribute	Specification
Selection	$\$ au_1 > R \quad au_2\$$
SLA	< 48 hours to desk compliance analyst

Table 13 – continued

Attribute	Specification
Investigator	Desk compliance analyst (L4–L5)
Workflow	Standard investigation. Ranked list within tier by R .
Triage	Analyst can escalate to Tier 1 if new evidence emerges.

8.3 Tier 3 — Automated

Attribute	Specification
Selection	$R < au_2$
Action	Auto-close. No human review.
Audit	Full feature vector, model version, SHAP values, and threshold parameters persisted for 7 years.
Random Audit	2% weekly sample forced to junior analyst review. Results feed back into label set.
Dispute	Traders can request manual review via compliance portal. Review is logged but does not trigger model retraining unless pattern emerges.

9. Evaluation & Monitoring

9.1 Primary Metrics

Metric	Definition	Target
Precision@K	Of top K alerts by R , % confirmed positives	$> 25\%$ (baseline: $\sim 5\%$ without prioritization)
Investigation Yield	Confirmed violations / total alerts in tier	Tier 1 $> 5\times$ Tier 2 $> 10\times$ Tier 3
Regulatory Miss Rate	Violations detected by regulators/whistleblowers that system missed	< 1 per year
Mean Time to Detection (MTTD)	Hours from trade execution to Tier 1 alert	< 4 hours
Tier 3 Audit Yield	Confirmed violations found in Tier 3 random audit	$< 2\%$ of audited alerts

9.2 Model Monitoring

Check	Frequency	Threshold	Action
PSI (Population Stability Index)	Daily	PSI > 0.25 on R distribution	Trigger retraining
Feature Drift	Daily	Any feature PSI > 0.2	Alert data engineering

Table 16 – continued

Check	Frequency	Threshold	Action
SHAP Drift	Weekly	Top 5 feature importances shift > 20%	Architecture review
Latency	Real-time	p99 > 50ms	Page on-call engineer
Throughput	Real-time	Queue depth > 100 alerts	Scale inference replicas
Label Flow	Weekly	< 5 new confirmed positives / week	Investigate data pipeline

9.3 Regulatory Miss Protocol

If a violation is detected by a regulator or whistleblower that the system missed:

1. **Immediate:** Extract full feature history for the missed case.
2. **Within 24 hours:** Root cause analysis — was it a model error, feature missingness, or threshold misalignment?
3. **Within 48 hours:** Emergency retrain if root cause is model-related.
4. **Within 1 week:** Report to Model Risk Committee with remediation plan.
5. **Quarterly:** Aggregate miss analysis presented to CRO and CCO.

10. Governance, Risk & Compliance (GRC)

10.1 Model Risk Management

This system is classified as a **high-risk model** under SR 11-7 (US) / SS1/13 (UK) / EU AI Act (High-Risk AI System).

Required Documentation:

- Model Concept Paper (this document)
- Model Development Document (MDD)
- Model Validation Report (independent validation)
- Ongoing Monitoring Report (quarterly)
- Model Inventory Entry (permanent)

Independent Validation:

- Must be performed by a team with no reporting line to Compliance Technology.
- Validates: data quality, methodology appropriateness, code review, stress testing, bias audit.

10.2 Bias & Fairness

Disparate Impact Audit (Quarterly):

- Test whether Tier 1 rate differs significantly across:

- Trading desks (equities vs. fixed income vs. FX)
- Geographic regions
- Instrument types (small-cap vs. large-cap)
- Trader demographics (tenure, role level)
- **Metric:** Adverse impact ratio (Tier 1 rate for group A / Tier 1 rate for group B). Flag if < 0.8 for any pair.
- **Remediation:** If bias detected, add fairness constraints to Stage 1 or remove proxy features (e.g., desk ID).

10.3 Audit Trail

Every tier assignment decision must be reproducible for 7 years:

- Alert ID and timestamp
- Complete feature vector (JSON)
- Model version strings for Stage 1, Stage 2, and composite logic
- Threshold values (\$ au_1, au_2, lpha\$) at time of assignment
- SHAP values for top 10 features
- Investigator outcome (if applicable)
- Regulatory action (if applicable)

10.4 Human-in-the-Loop Requirements

- **Tier 1:** Model recommends; human decides. No auto-escalation to regulators.
- **Tier 2:** Model recommends; human investigates. Human can override tier assignment with documented rationale.
- **Tier 3:** Model decides; human audits sample. No individual human review for closure.
- **Override Logging:** All human overrides are logged and analyzed monthly for model improvement.

10.5 Regulatory Reporting Interface

The system must expose APIs for:

- **SEC:** Real-time alert data for MRT (Market Abuse Regulatory Team) inquiries.
- **FCA:** Bulk export of Tier 1 alert histories for suspicious transaction reporting.
- **Internal Audit:** Read-only access to all audit logs, model versions, and SHAP explanations.

11. Implementation Roadmap

Phase 1: Foundation (Weeks 1–8)

Objective: Build data infrastructure and baseline model.

Week	Deliverable
1-2	Feature store architecture design and data source integration
3-4	Feature engineering pipeline (batch + real-time)
5-6	Baseline Stage 1 model (two-step PU method) on 12-month historical data
7-8	Validation against confirmed positive set; baseline Precision@K measurement

Milestone: Feature store operational. Baseline model achieves Precision@K > 15% on historical data.

Phase 2: Model Development (Weeks 9–16)

Objective: Build full 3-stage stack and optimize tiers.

Week	Deliverable
9-10	Stage 1 nnPU implementation and hyperparameter tuning
11-12	Stage 2 severity model (transfer learning from public enforcement data if needed)
13-14	Stage 3 composite score and cost-optimal threshold optimization
15-16	End-to-end pipeline integration; SHAP explainability layer

Milestone: Full 3-stage model operational in shadow mode. Precision@K > 20% on validation set.

Phase 3: Operational Integration (Weeks 17–24)

Objective: Deploy with human oversight and measure live performance.

Week	Deliverable
17-18	Shadow deployment: model scores all alerts but does not route. Parallel investigation of model Tier 1 vs. existing rule-based Tier 1.
19-20	A/B test analysis; model refinement based on live data
21-22	Graduated rollout: auto-route Tier 3 only; human oversight on Tier 1/2
23-24	Full production deployment; investigator training on SHAP narratives

Milestone: Live system routing 100% of alerts. Tier 3 auto-closure active. Investigator yield measured weekly.

Phase 4: Optimization & Governance (Weeks 25–36)

Objective: Harden monitoring, governance, and continuous improvement.

Week	Deliverable
25-28	Bias audit; disparate impact remediation; model risk documentation
29-32	Independent validation; regulatory pre-notification

Table 20 – continued

Week	Deliverable
33–36	Automated retraining pipeline; drift monitoring dashboard; quarterly review cycle

Milestone: Model Risk Committee sign-off. System enters steady-state operations.

12. Appendices

Appendix A: Glossary

Term	Definition
MAD	Market Abuse Detection
PU Learning	Positive-Unlabeled learning; training with confirmed positives and unlabeled samples
nnPU	Non-negative unbiased risk estimator for PU learning
SHAP	SHapley Additive exPlanations; model-agnostic feature attribution
PSI	Population Stability Index; measures distribution drift
NFA	No Further Action; investigator closure without violation
HHI	Herfindahl-Hirschman Index; concentration measure
MRT	Market Abuse Regulatory Team (SEC)
SAR	Suspicious Activity Report

Appendix B: Data Retention & Privacy

- **Alert data:** 7 years (regulatory requirement).
- **Feature vectors:** 7 years, encrypted at rest.
- **SHAP explanations:** 7 years, linked to alert ID.
- **Trader PII:** Minimized in model features. Tenure and role are used; name, age, gender, ethnicity are **excluded** from all model inputs.
- **Communication metadata:** Aggregated graph centrality only; no message content or recipient identities in features.

Appendix C: Risk Register

Risk	Likelihood	Impact	Mitigation
PU learning instability (nnPU divergence)	Medium	High	Fallback to two-step method; gradient clipping; validation monitoring
Severe model drift (market regime change)	Low	Critical	Weekly PSI monitoring; emergency retraining protocol; human override capability

Table 22 – continued

Risk	Likelihood	Impact	Mitigation
Regulatory miss due to Tier 3 auto-closure	Low	Critical	2% random audit; quarterly regulatory miss analysis; feedback loop to labels
Investigator resistance to ML recommendations	Medium	Medium	SHAP narratives; override logging; monthly yield reporting to demonstrate value
Feature pipeline failure (stale data)	Medium	High	Real-time freshness checks; circuit breakers to rule-based fallback if features > 1 hour stale
Bias audit failure	Low	High	Quarterly disparate impact testing; fairness constraints; proxy feature removal

Appendix D: Fallback Procedures

If model inference fails or latency exceeds 50ms:

1. Circuit breaker activates after 3 consecutive failures.
2. System falls back to **upstream anomaly score percentile** as temporary prioritization.
3. All alerts default to Tier 2 (no auto-closure) until model recovers.
4. On-call engineer paged automatically.
5. Post-incident review within 24 hours.

If feature store is unavailable:

1. Use last-known feature values if < 1 hour old.
2. If > 1 hour old, reject alert scoring and queue for batch reprocessing.
3. Upstream MAD engine continues to generate alerts; they are held in queue.

Document Control:

- **Author:** Compliance Data Science
- **Reviewer:** Model Risk Management, Legal, Compliance Operations
- **Approver:** Chief Compliance Officer, Chief Risk Officer
- **Next Review Date:** 2026-09-27

END OF SPECIFICATION